

AP CALCULUS AB - UNIT 1

1.2 Defining Limits and Using Limit Notation



Lesson Overview

- Interpret every symbol in finite, one-sided, and infinite limit notation.
- Translate among words, graphs, tables, and limit statements.
- Use one-sided agreement to determine whether a two-sided limit exists.
- Distinguish a limit, a point value, an infinite limit, and a statement that the limit does not exist.

Key Knowledge and Vocabulary

Nearby behavior. $\lim_{x \rightarrow c} f(x) = L$ describes $f(x)$ for x close to but not necessarily equal to c .

One-sided notation. $x \rightarrow c^-$ means inputs below c ; $x \rightarrow c^+$ means inputs above c .

Agreement theorem. A finite two-sided limit equals L exactly when both one-sided limits equal L .

DNE. Use DNE when the required finite agreement fails; do not use “undefined” as a substitute.

Infinite limits. $\lim_{x \rightarrow c^-} f(x) = +\infty$ means values grow without bound positively from the left; $+\infty$ is not a real output value.

Explanation and Strategy

Step 1. Read a limit statement in the order: expression, input target, input direction, output destination.

Step 2. Always separate $\lim_{x \rightarrow c} f(x)$ from $f(c)$. Either may exist without the other.

Step 3. Before writing a two-sided limit, record the left- and right-hand limits. If they differ, the two-sided limit does not exist.

Solved Examples

Worked Example: Words to symbols

Write: “As x approaches -3 from values greater than -3 , $p(x)$ decreases without bound.”

Solution. $\lim_{x \rightarrow -3^+} p(x) = -\infty$.

Worked Example: Point value versus limit

A graph approaches 6 from both sides at $x = 2$, has an open circle at $(2, 6)$, and a filled point at $(2, -1)$. State the relevant facts.

Solution. $\lim_{x \rightarrow 2^-} f(x) = 6$, $\lim_{x \rightarrow 2^+} f(x) = 6$, and therefore $\lim_{x \rightarrow 2} f(x) = 6$, while $f(2) = -1$.

Worked Example: Failure by disagreement

Suppose $\lim_{x \rightarrow 4^-} h(x) = 2$ and $\lim_{x \rightarrow 4^+} h(x) = 7$. What can be concluded?

Solution. The two-sided limit $\lim_{x \rightarrow 4} h(x)$ does not exist because the one-sided limits are finite but unequal. No conclusion about $h(4)$ follows.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Translate $\lim_{x \rightarrow 5} f(x) = 9$ into a complete sentence.

2. Write notation for: “ $q(x)$ approaches -4 as x approaches 1 from the left.”

3. Write notation for: “ $r(x)$ grows without bound as x approaches 0 from the right.”

4. What does the superscript $+$ in $x \rightarrow c^+$ describe?

Medium Level

Apply the idea in one or two connected steps.

5. Given $\lim_{x \rightarrow 3^-} f(x) = 8$ and $\lim_{x \rightarrow 3^+} f(x) = 8$, state the two-sided limit.

6. Given $\lim_{x \rightarrow -1^-} g(x) = 0$ and $\lim_{x \rightarrow -1^+} g(x) = 5$, state the two-sided conclusion.

7. A function is not defined at $x = 6$, but nearby values approach 11 . Write two correct mathematical statements.

8. Explain why $\lim_{x \rightarrow 2} f(x) = 4$ does not imply $f(2) = 4$.

Hard Level

Connect representations, conditions, and justification.

9. Suppose $\lim_{x \rightarrow a} f(x) = L$. Must $f(a)$ exist? Must L be finite? Answer for the stated notation.

10. Write a complete set of one-sided statements describing a vertical asymptote at $x = -2$ with both branches rising.

11. Can $\lim_{x \rightarrow 4} f(x) = 3$ and $f(4) = 100$ both be true? Justify.

12. Correct the statement: “Because $f(1)$ is undefined, $\lim_{x \rightarrow 1} f(x)$ is DNE.”

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Assume $\lim_{x \rightarrow c^-} f(x) = L$ and $\lim_{x \rightarrow c} f(x) = M$, where both are finite. Prove that $L = M$.

14. Write limit notation that distinguishes these two behaviors near $x = 0$: (i) outputs alternate between -1 and 1 ; (ii) outputs become arbitrarily large in magnitude with opposite signs on opposite sides.

15. If $\lim_{x \rightarrow c} f(x) = L$, determine the limit of $f(x)$ along any sequence $x_n \neq c$ with $x_n \rightarrow c$.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Construct a function for which $\lim_{x \rightarrow 0^-} f(x) = 2$, $\lim_{x \rightarrow 0^+} f(x) = 2$, and $f(0)$ is undefined.

17. Construct a function satisfying $f(0) = 7$, $\lim_{x \rightarrow 0^-} f(x) = -1$, and $\lim_{x \rightarrow 0^+} f(x) = 3$. State the two-sided limit.

AP-Style Practice**Multiple-Choice Practice – No Calculator**

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. If $\lim_{x \rightarrow 2^-} f(x) = 5$ and $\lim_{x \rightarrow 2^+} f(x) = 5$, which must be true?

- (A) $f(2) = 5$ (B) $f(2)$ exists
(C) $\lim_{x \rightarrow 2} f(x) = 5$ (D) f is differentiable at 2

2. Which notation means that $g(x)$ decreases without bound as x approaches 4 through values less than 4?

- (A) $\lim_{x \rightarrow 4^-} g(x) = -\infty$ (B) $\lim_{x \rightarrow 4^+} g(x) = -\infty$
(C) $\lim_{x \rightarrow -4^-} g(x) = +\infty$ (D) $g(4) = -\infty$

3. Which situation alone is sufficient to conclude that $\lim_{x \rightarrow c} f(x)$ does not exist?

- (A) $f(c)$ is undefined. (B) $f(c) \neq 0$.
(C) The finite one-sided limits are unequal. (D) The graph has a corner at c .

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator table produces the data below. Which notation best describes the behavior?

x	1.99	1.999	2.001	2.01
$g(x)$	-48.7	-498.7	501.3	51.3

- (A) $\lim_{x \rightarrow 2} g(x) = 0$ (B) $\lim_{x \rightarrow 2^-} g(x) = -\infty$ and $\lim_{x \rightarrow 2^+} g(x) = +\infty$
 (C) $\lim_{x \rightarrow 2} g(x) = +\infty$ (D) $g(2) = 0$

5. Near $x = 1.5$, a graphing calculator shows $h(x)$ approaching 3.000 from both sides, but the calculator reports “undefined” at $x = 1.5$. Which conclusion is best?

- (A) $h(1.5) = 3$ (B) $\lim_{x \rightarrow 1.5} h(x) = 3$
 (C) The limit is undefined. (D) The limit is infinite.

Free-Response Practice – No Calculator

A function f is described near $x = 3$ by $\lim_{x \rightarrow 3^-} f(x) = 4$, $\lim_{x \rightarrow 3^+} f(x) = 4$, and $f(3) = -2$.

(a) State $\lim_{x \rightarrow 3} f(x)$ and justify. [2 points]

(b) Is f continuous at $x = 3$? Explain using the information given. [2 points]

(c) Write a sentence explaining why the point value does not alter the answer to part (a). [2 points]

Free-Response Practice – Calculator Required

A calculator investigation of a function q near $x = -1$ gives $q(-1.01) = 99.4$, $q(-1.001) = 999.4$, $q(-0.999) = -1000.6$, and $q(-0.99) = -100.6$.

(a) Write the two one-sided infinite-limit statements supported by the data. [2 points]

(b) Does $\lim_{x \rightarrow -1} q(x)$ exist as a finite limit? Explain. [2 points]

(c) What vertical asymptote is suggested? [1 points]